MATH 152 **QUIZ 3** Name _____

Mrs. Bonny Tighe 25 points

6.5,7.1 TA_____ Wed. 9/19/18

1. Find the average value of $f(x) = \dfrac{\sec^2 x}{1 + \tan x}$ on $[0, \pi/6]$

2. The linear density in a rod 2 m long is $\dfrac{10x}{\sqrt{x^2+1}}$ kg/m where x is measured in meters from one end of the rod. Find the average density of the rod.

3. Find the numbers *b* such that the average value of $f(x) = 3x^2 - 12x + 13$ on the interval [0, *b*] is equal to 5.

4. Evaluate the integrals. a) $\int x(\csc^2 x)dx$

b) $\int \tan^{-1}\left(\frac{1}{x}\right) dx$

c) $\int e^{3x} \sin 2x \; dx$

d) $\int t^{2} 5^{-2t} \, dt$

MATH 152 **QUIZ 3A** Name _____

Mrs. Bonny Tighe 25 points

6.5, 7.1 TA_____ Wed 9/18/2019

1. Find the average value of $f(x) = \dfrac{3^x}{1+3^{2x}}$ on [0,1]

2. A cup of coffee has temperature 95° C and takes 30 minutes to cool to 61° C in a room with temperature 20° C. By Newton's Law of Cooling, the temperature of the coffee after t minutes is $T(t) = 20 + 75e^{-0.5t}$. What is the average temperature of the coffee during the first half hour?

3. Evaluate the integrals. a) $\int \tan^{-1} t \; dt$

b) $\int e^{2x} \sin(3x)\, dx$

c) $\int \cos\sqrt{x}\, dx$

d) $\int x^4 \ln x \, dx$